

HARSH PATEL

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SUMMARY

Robotics & Autonomy M.S. candidate targeting entry-level Robotics Software and Autonomy roles, with hands-on experience building ROS / ROS 2, C++, and Python systems for mobile robot navigation, localization, SLAM, path planning, and manipulation. Built and validated TurtleBot3, OpenManipulator-X, EKF/MCL, A*/D* Lite, MoveIt, and Gazebo/RViz projects across simulation and hardware, backed by manufacturing and process-engineering experience for hardware-aware development and real-world system validation.

EDUCATION

Drexel University — M.S. Robotics and Autonomy

Philadelphia, PA | Expected Sep 2026

Gujarat Technological University — B.E., Mechanical Engineering

Ahmedabad, India | Jul 2016 – Aug 2020

EXPERIENCE

DwyerOmega Engineering — Manufacturing Engineering Co-op

Bridgeport, NJ | Oct 2025 – Mar 2026

- Created and maintained SolidWorks parts, assemblies, and drawings for component fit checks, manufacturability reviews, engineering-change documentation, and tooling/fixture support.
- Supported manufacturing engineering and automation-improvement initiatives for sensor and instrumentation production workflows, improving documentation consistency and production-floor execution.
- Authored standardized work instructions, engineering documentation, and Excel-based tools to reduce manual lookup errors, assembly variation, and process documentation inconsistencies.

Flowell Water Pumps — Senior Process Engineer

Ahmedabad, India | Dec 2020 – Aug 2024

- Drove CAD-based process improvements with mechanical, electrical, QA, and production teams, contributing to a 15% reduction in manufacturing cost.
- Led commissioning, testing, and optimization of high-efficiency centrifugal pump assemblies, improving hydraulic performance and reliability across production lines.
- Executed prototype performance testing and translated test data into design and process recommendations to improve pump reliability.

TECHNICAL SKILLS

Programming and ROS: Python, C++, C, MATLAB, Bash, ROS 2, ROS (Noetic), TF/TF2, MoveIt, URDF, Nodes/Topics, Launch Files, rosbag

Planning & Control: A*, D* Lite, RRTConnect, PRM, KPIECE, Trajectory Generation, Jacobian-Based IK, PID, ADRC

Perception & Vision: OpenCV, Stereo Calibration, SIFT/FLANN Matching, Harris Corners, Edge Detection

Hardware & CAD: Arduino, TurtleBot3, OpenManipulator-X, Sensor Integration, PLC (Mitsubishi GX Works2), SolidWorks

PROJECTS

Autonomous Patrol Robot — Behavior Trees & Marker

- Building a C++ patrol behavior with BehaviorTree.CPP v4 that sequences patrol, battery-recovery, and marker-response branches, commanding a robot through the Nav2 stack in simulation.
- Developed a Python perception node using OpenCV ArUco marker detection that publishes detection state on a ROS 2 topic to trigger behavior-tree transitions.
- Integrating the perception topic into the C++ behavior tree for reactive, marker-driven patrol behavior (current milestone).

Autonomous Warehouse Robot Path Planning

- Implemented a D* Lite graph-search planner in ROS/Gazebo for dynamic warehouse path-planning, enabling obstacle-aware replanning when new obstacles appeared during navigation.
- Applied dynamic obstacle avoidance using Artificial Potential Field (APF) path planning.
- Validated goal-to-goal robot routes in Gazebo and RViz, confirming planner behavior across changing obstacle layouts, and containerized the ROS/Gazebo environment with Docker for a reproducible one-command setup.

OpenManipulator-X Manipulation & Trajectory Planning

- Validated URDF-based forward kinematics and damped-least-squares inverse kinematics against MoveIt end-effector poses to confirm manipulator model accuracy.
- Benchmarked RRTConnect, PRM, and KPIECE planners using planning time, final pose error, and tracking error to compare motion-planning performance.
- Developed ROS/Python nodes for joint-space, Cartesian, waypoint, and quintic-polynomial trajectory execution, enabling repeatable manipulator motion testing.

Computer Vision & Stereo Calibration

- Implemented stereo camera calibration, Harris-corner detection, edge detection, and SIFT/FLANN feature matching in OpenCV to support visual perception and scene reconstruction.
- Estimated scene depth from calibrated stereo imagery and evaluated measurement accuracy against known image geometry.

ADRC-Controlled Sensor System

- Deployed an **Active Disturbance Rejection Controller** on an Arduino LED–photoresistor platform, using C-based sensor/actuator logic to improve tracking under disturbance conditions.
- Modeled and validated system response in MATLAB/Simulink to compare controller behavior, tracking stability, and disturbance rejection performance.

CERTIFICATIONS

ROS 2 & ROS (Noetic/Melodic/Kinetic) · MATLAB & Simulink Onramp (MathWorks) · Mitsubishi GX Works2 PLC (Udemy)